

A disproof of Conjecture 00000007661

Q-geometric ratios of q-hypergeometric integer series

1 The conjecture

Conjecture 00000007661 reads:

Definition: Let $0 < |q| < 1$; a power series $F(z) = \sum a_n z^n$ is of *q-hypergeometric type* if there are rational functions $R(z), S(z)$ with $F(qz) = R(z)F(z) + S(z)$; the coefficient ratios a_{n+1}/a_n are the *Q-geometric ratios*. *Conjecture:* $F(z) \in \mathbb{Z}[[z]]$ is of q-hypergeometric type with convergence radius 1 if and only if its Q-geometric ratio sequence converges to 1 at rate $O(q^n)$, and this rate cannot be improved to $o(q^n)$ unless F is a polynomial.

Fix $q \in \mathbb{C}$ with $0 < |q| < 1$ (the Lean proof works for every such q , so it refutes both the reading “for every such q ” and the reading “for some such q ”). Write $L(F)$ for “ F is of q-hypergeometric type and has radius of convergence 1”, $O(F)$ for “ $a_{n+1}/a_n \rightarrow 1$ and $a_{n+1}/a_n - 1 = O(q^n)$ ”, and $o(F)$ for “ $a_{n+1}/a_n - 1 = o(q^n)$ ”. We refute the following three readings:

- (A) the equivalence $L(F) \iff O(F)$ for all $F \in \mathbb{Z}[[z]]$;
- (B) the sharpness clause $o(F) \Rightarrow F$ is a polynomial, even restricted to F with $L(F)$;
- (C) the equivalence $L(F) \iff (O(F) \wedge (o(F) \Rightarrow F \text{ is a polynomial}))$, in which the sharpness clause is part of the right-hand side.

The statement read as (A) and (B) together is false because (A) is false (and also because (B) is false).

2 Definitions and conventions

- $F = \sum a_n z^n \in \mathbb{Z}[[z]]$ is evaluated as the complex sum $F(z) = \sum_{n \geq 0} a_n z^n$ (Lean: `toFun`, `atsum`); the radius of convergence is the radius of the complex power series $\sum a_n z^n$ (Lean: `radius`, the radius of `FormalMultilinearSeries.ofScalars`).
- A rational function is A/B with $A, B \in \mathbb{C}[z]$, $B \neq 0$. Analytic reading (`QHypergeometricType`): there are $A, B, C, D \in \mathbb{C}[z]$ with $B, D \neq 0$ such that $F(qz) = \frac{A(z)}{B(z)}F(z) + \frac{C(z)}{D(z)}$ for every $|z| < 1$ with $B(z) \neq 0 \neq D(z)$. Formal reading (`QHypergeometricTypeFormal`): $BD \cdot F(qz) = AD \cdot F + BC$ in $\mathbb{C}[[z]]$, where $F(qz)$ is the series $\sum q^n a_n z^n$ (Mathlib `rescale q`).
- The Q-geometric ratio is a_{n+1}/a_n , computed in $\mathbb{R}$ (Lean: `ratio`). Both examples below have all $a_n \neq 0$, so no convention for division by zero is used.

- $O(q^n)$ and $o(q^n)$ are Mathlib's `IsBig0` and `IsLittle0` along $n \rightarrow \infty$; they compare $|a_{n+1}/a_n - 1|$ with $|q|^n$.
- “ F is a polynomial” means F is the image of some $p \in \mathbb{Z}[z]$ (`IsPoly`).

3 The counterexamples

Let $F_1 = \sum_{n \geq 0} z^n = 1/(1 - z)$ and $F_2 = \sum_{n \geq 0} (n + 1)z^n = 1/(1 - z)^2$, both in $\mathbb{Z}[[z]]$.

Lemma 1. *For $|z| < 1$, $F_1(z) = (1 - z)^{-1}$ and $F_2(z) = (1 - z)^{-2}$. Hence $F_1(qz) = \frac{1-z}{1-qz} F_1(z)$ and $F_2(qz) = \frac{(1-z)^2}{(1-qz)^2} F_2(z)$ on the unit disk, and both are of q -hypergeometric type ($S = 0$) in the analytic and the formal sense.*

Proof. The geometric series gives F_1 ; $\sum n z^n = z/(1 - z)^2$ (Mathlib `hasSum_coe_mul_geometric_of_norm_lt_one`) plus $\sum z^n$ gives F_2 . Since $|qz| < 1$, the same formulas hold at qz , and $1 - z \neq 0$. Formally, $(1 - z)F_1 = 1$ and $(1 - z)^2 F_2 = 1$ in $\mathbb{C}[[z]]$ (Mathlib `mk_one_mul_one_sub_eq_one`, `mk_add_choose_mul_one_sub_pow_eq_one`); applying the ring homomorphism $z \mapsto qz$ gives $(1 - qz)F_1(qz) = 1$ and $(1 - qz)^2 F_2(qz) = 1$, so both sides of the cleared identities equal 1. (Lean: `qHyp_F1`, `qHyp_F2`, `qHypFormal_F1`, `qHypFormal_F2`.) $\square$

Lemma 2. *Both F_1 and F_2 have radius of convergence 1.*

Proof. $|a_{n+1}|/|a_n|$ equals 1 for F_1 and $(n + 2)/(n + 1) = 1 + 1/(n + 1) \rightarrow 1$ for F_2 ; by the ratio test (Mathlib `ofScalars_radius_eq_inv_of_tendsto`) the radius is 1. (Lean: `radius_F1`, `radius_F2`.) $\square$

Lemma 3. *(i) For F_2 , $a_{n+1}/a_n - 1 = 1/(n + 1)$, which is not $O(q^n)$. (ii) For F_1 , $a_{n+1}/a_n = 1$, so $O(F_1)$ and $o(F_1)$ hold; F_1 is not a polynomial.*

Proof. (i) If $1/(n + 1) \leq c|q|^n$ for all large n , then $1 \leq c(n + 1)|q|^n$, but $(n + 1)|q|^n \rightarrow 0$ (Mathlib `tendsto_self_mul_const_pow_of_lt_one`). (ii) The zero sequence is o of anything. If $F_1 = p$ for a polynomial p , the coefficient of $z^{\deg p + 1}$ would be both 1 and 0. (Lean: `not_rateBig0_F2`, `rateBig0_F1`, `rateLittle0_F1`, `not_isPoly_F1`.) $\square$

Theorem 4. *For every $q \in \mathbb{C}$ with $0 < |q| < 1$, readings (A), (B), (C) are all false, with either reading of “ q -hypergeometric type”.*

Proof. (A): $L(F_2)$ holds but $O(F_2)$ fails. (B): $L(F_1)$ and $o(F_1)$ hold but F_1 is not a polynomial. (C): $L(F_1)$ holds but the right-hand side fails at F_1 , since $o(F_1)$ holds and F_1 is not a polynomial. $\square$

4 Lean formalization

File `lean/Conjecture7661/Basic.lean` (namespace `C7661`, Lean 4 + Mathlib, only `import Mathlib`). Main theorems:

- `conjecture_false` (`q : C`) (`_hq0 : 0 < norm q`) (`hq1 : norm q < 1`) (ASCII rendering): the facts about F_2 and F_1 above, and the negations of (A), (B) (restricted to L) and (C), with the analytic definition `QHypergeometricType`.
- `conjecture_false_formal`: the negations of (A), (B), (C) with the formal definition `QHypergeometricTypeFormal`.

Axiom audit: `#print axioms` for both theorems reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`.

Remark 5. We do not refute any reading in which the right-hand side does not imply $a_{n+1}/a_n - 1 = O(|q|^n)$ and the sharpness clause is dropped; we know of no such reading of the text.